



# Lucas Shuo Han

韩朔

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Peking University, National School of Development  
B.S. in Economics & Mathematics, Expected 2028

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## PROJECTS

### Desktop Intelligent Micro-Ecosystem

C, Embedded Systems

- Designed and programmed the control system for a self-sustaining micro-ecosystem, integrating DHT11 humidity sensing, voice recognition module, and relay control to maintain 40-60% RH and respond to Mandarin voice commands.
- Responsible for hardware selection, circuit integration, and firmware development that ties the sensing and actuation together.
- [lucas-shuo-han.github.io/project1.html](https://lucas-shuo-han.github.io/project1.html)

### Wren Mating System Modeling

Python, Game Theory, Agent-Based Modeling

- Developed a grid-based agent model to simulate mating system evolution using an energy budget abstraction—compressing complex reproductive behaviors (mate acceptance, nest abandonment, parental care) into a single continuous variable.
- Implemented Bayesian social learning for strategy updates and designed the decision-making architecture with clean separation of concerns for extensibility.
- [github.com/lucas-shuo-han/BreedShare.git](https://github.com/lucas-shuo-han/BreedShare.git)

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## RESEARCH EXPERIENCE

### Research Assistant, Dandan Zhang Lab

Dec 2025 – Present

*National School of Development, Peking University*

- Examining how extreme weather affects delivery workers' labor supply decisions when platforms lack information on individual reservation prices.
- Using empirical methods to estimate subsidy effects on voluntary participation, aiming to inform smarter dispatch algorithms and more flexible incentive design.

### Research Assistant, Jackson Champer Lab

Sep 2025 – Dec 2025

*School of Life Sciences, Peking University*

- Assisted with routine lab maintenance and data collection; gained exposure to frontline scientific research workflows and operations.

### Library Volunteer

2024 – Present

*Peking University Library*

- Rotated through shelving, interlibrary loan, and special collections; gained comprehensive understanding of library operations and information systems management.

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## TECHNICAL SKILLS

Languages: Python, C

Data: Stata, Pandas

Tools: Git, Shell, Docker, Linux

Languages: Chinese (native), English (TOEFL R6 L6 W6 S5.5)

**HONORS & AWARDS**

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|---------------------------------------------------------|----------|
| Chinese Mathematics Competitions (Beijing), First Prize | Nov 2025 |
| NSD Summer Social Practice Outstanding Participant      | Aug 2025 |
| Peking University Weiming Reading Star                  | May 2025 |
| Peking University Freshman Scholarship, Third Class     | Dec 2024 |
| China National Biology Olympiad, Second Prize           | Aug 2023 |